

MANOJ KUMAR VIRIGINENI

Sr Data Engineer/ Cloud Engineer

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Executive Summary

Senior Data Engineer with 9+ years of experience in cloud data warehousing and big data processing. Expertise in Snowflake, AWS, Databricks and Apache Spark with proven success in architecting scalable data pipelines, implementing ETL workflows and managing enterprise data platforms. Delivered significant cost optimization and performance improvements through automated resource management and infrastructure optimization.

Key Achievements

- Facilitated development of an AI-Powered mortgage analysis platform integrating Amazon Bedrock with Snowflake, achieving a 70% reduction in manual analysis workload and 95% accuracy in risk assessments while accelerating loan approvals from days to hours.
- Delivered over \$420,000 in annual savings through strategic data warehouse optimization and automated resource management, reducing operational costs by 35% and improving performance by 25% across enterprise data systems.
- Successfully executed end-to-end migration of over 1,000 tables and 2 TB of data from SQL Server to Snowflake using AWS DMS (Data Migration Service) completing the project ahead of critical deadline while enhancing cloud analytics capabilities.
- Enhanced security posture by deploying role-based access controls (RBAC) and optimizing permissions workflows, minimizing breach risk while ensuring compliance and operational efficiency.
- Reduced costs by implementing auto-suspend, dynamic cluster sizing, and automated scaling, ensuring optimal cloud data infrastructure performance.
- Earned multiple advanced cloud certifications demonstrating expertise across modern data platforms including [SnowPro Core Certification \(Snowflake\)](#), [DBT Certified Developer \(dbt Labs\)](#), and [AWS Certified AI Practitioner \(Amazon Web Services\)](#).

Professional Experience

Xpanse

Frisco, TX

Sr Data Engineer/Snow Admin

01/2024 - Present

- Engineered enterprise-wide cost optimization framework reducing Snowflake infrastructure expenses by \$125K annually (28% cost reduction) through automated warehouse scaling, resource monitoring dashboards, and intelligent query optimization across 8 departments and 50+ data pipelines.
- Architected AI-Powered mortgage analysis platform integrating Amazon Bedrock with Snowflake and streamlit, reducing manual analysis workload by 70% and accelerating loan approvals from 3-5 days to under 2 hours while achieving 95% accuracy in risk assessments for 50k+ monthly mortgage applications.
- Optimized Snowflake warehouse performance through dynamic sizing and query tuning, increasing query execution speed by 40% (from average 120 seconds to 72 seconds) while reducing compute costs by 22% and improving analyst productivity for 200+ users.
- Implemented dbt Core transformation framework replacing 200+ legacy SQL stored procedures, improving pipeline maintainability by 80%, reducing deployment time from 4 hours to 30 minutes, and enabling version-controlled, testable data transformations across 1,000+ tables.
- Designed comprehensive RBAC security architecture with 50+ data masking policies and role hierarchies protecting sensitive financial data across 1,000+ Snowflake tables, achieving zero security incidents, 100% SOC 2 compliance audit results, and meeting all regulatory requirements.
- Deployed Snowflake Time Travel and Fail-safe capabilities enabling 90-day data recovery window, successfully recovering from 15+ accidental data modification incidents with zero data loss and average recovery time of 12 minutes.
- Automated database provisioning workflows using Python and Snowflake stored procedures, reducing schema deployment time by 65% (from 4 hours to 90 minutes) and eliminating 80% of manual deployment errors across 30+ monthly releases.
- Established automated data quality framework integrating dbt tests with Great Expectations, detecting and preventing 300+ data anomalies monthly, improving data trust scores from 72% to 94%, and reducing downstream analytics errors by 60%.
- Orchestrated 24/7 production support for 50+ mission-critical data pipelines processing \$2B+ monthly mortgage volume, maintaining 99.8% uptime SLA, resolving P1 incidents within 30-minute target, and documenting 100+ issue patterns for automated prevention.
- Engineered secure multi-tenant data pipelines processing 500GB daily sensitive mortgage data with PII encryption, automated audit logging, field-level masking, and real-time compliance monitoring ensuring GLBA and SOX regulatory adherence.
- Developed prompt engineering frameworks for LLM-based mortgage document summarization and loan eligibility reasoning, leveraging contextual embeddings from Snowflake Cortex and AWS Bedrock models.
- Designed and implemented Redpoint campaign data integrations with Snowflake Gold Layer datasets to support Retail Refill Reminder (RRR) and GI campaigns, enabling real-time eligibility segmentation and personalized outreach.
- Designed and implemented Ataccama ONE 16.2-based Data Quality framework integrated with Snowflake data warehouse to monitor, validate, and maintain critical data elements (CDEs) across multiple domains.

Professional Experience

Freedom Mortgage

New Jersey, NY

Data Engineer

06/2022 - 12/2023

- Executed end-to-end enterprise migration of 1,000+ database tables (2TB data, 500+ stored procedures) from on-premise SQL Server to Snowflake using AWS DMS, completing project 2 weeks ahead of schedule with 99.9% data integrity verification and zero business downtime across 12 production systems.
- Engineered real-time data processing architecture using Snowpipe and AWS Lambda reducing data latency from 4 hours to 15 minutes, enabling near real-time mortgage decisioning for 50K+ daily applications, and improving operational efficiency by 85% for downstream analytics teams.
- Architected comprehensive Snowflake data warehouse integrated with 12 AWS services (S3, Lambda, SNS, SQS, Glue, IAM, CloudWatch, DMS) processing \$2B+ monthly mortgage volume, achieving 99.95% pipeline reliability, sub-second query response for executive dashboards, and 35% cost optimization through automated resource management.
- Implemented enterprise-grade data quality monitoring system using Atacama across 200+ data sources, reducing data anomalies by 60% (from 500 to 200 monthly incidents), preventing \$1.2M in potential compliance penalties, and improving data trust scores from 68% to 92% among business stakeholders.
- Developed and orchestrated 200+ SnapLogic ETL pipelines with Control-M scheduling processing 15 million daily records, improving data refresh frequency by 50% (from 4 hours to 2 hours), reducing manual intervention by 75%, and automating 85% of previously manual data validation workflows.
- Designed batch and streaming data integration frameworks handling 3TB daily mortgage data across AWS infrastructure, achieving 99.8% SLA compliance, reducing processing time by 45%, and enabling real-time risk scoring for underwriting decisions.
- Generated 50+ complex regulatory reports (HMDA, QM/ATR, TRID) automating 70% of manual reporting workflows, ensuring 100% compliance with CFPB requirements, reducing report generation time from 3 days to 4 hours, and eliminating audit findings for 18 consecutive months.
- Extensive experience in creating batch & streaming data pipelines in AWS & Big Data Infrastructure.

Freedom Mortgage

New Jersey, NY

Data Modeler

08/2021 - 05/2022

- Designed and optimized 150+ dimensional data models for Snowflake using Erwin (Kimball methodology), improving query performance by 45% (from 180 seconds to 99 seconds average) and enabling sub-second response times for executive dashboards serving 200+ business users analyzing \$50B annual mortgage portfolio.
- Architected enterprise data warehouse supporting \$50B+ mortgage origination volume with 85+ fact and dimension tables, implementing star schema designs, slowly changing dimensions (SCD Type 2), and conformed dimensions reducing data redundancy by 40% and improving analytical flexibility for 30+ business units.
- Led data modeling standards initiative for Enterprise Data Architecture (EDA) team, establishing governance framework for 300+ tables, reducing modeling inconsistencies by 70%, accelerating new model delivery by 40% (from 2 weeks to 1-2 weeks), and improving cross-team collaboration efficiency.
- Collaborated with 8 cross-functional teams through 50+ JAD (Joint Application Design) sessions gathering requirements and delivering 25+ conceptual, logical, and physical data models supporting critical regulatory reporting with 100% compliance rate and zero audit findings across HMDA, TRID, and QM/ATR requirements.
- Implemented medallion architecture data lake (Bronze/Silver/Gold layers) organizing 500+ datasets across 3 zones, reducing data discovery time by 75% (from 3 days to 18 hours), improving data lineage documentation for 1,200+ data elements, and enabling self-service analytics for 150+ business analysts.
- Orchestrated 40+ weekly production deployments across 5 environments with 98% deployment success rate, implementing automated validation scripts reducing rollback incidents by 85%, and maintaining comprehensive change logs ensuring audit compliance and disaster recovery capabilities.
- Conducted comprehensive data profiling and quality assessments across 200+ source systems, identifying and resolving 300+ data quality issues, improving source data accuracy from 78% to 94%, and establishing automated data quality rules preventing downstream analytical errors.

Professional Experience

T-Mobile

San Francisco, United States

Big Data Engineer

08/2019 - 07/2021

- Developed enterprise-scale PySpark ETL pipelines processing 5TB daily telecom data (call records, network logs, customer interactions) across Azure Synapse Analytics and Databricks, improving processing speed by 60% (from 8 hours to 3.2 hours) and reducing infrastructure costs by \$180K annually through optimized Spark configurations.
- Designed Azure Data Lake medallion architecture (Bronze, Silver, Gold zones). Managed 800+ datasets. Cut data discovery by 75% (2 days to 12 hours). Boosted query performance by 50%. Enabled self-service for 500+ analysts and scientists.
- Engineered real-time streaming pipelines using Spark Structured Streaming and Azure Event Hubs processing 2 million transaction events per minute, enabling sub-second fraud detection, reducing customer churn by identifying at-risk customers 48 hours earlier, and improving campaign targeting ROI by 35%.
- Migrated 50+ legacy on-premise ETL workflows (DataStage, Informatica) to Azure cloud platform eliminating 12 physical servers, reducing maintenance overhead by 80% (from 40 hours to 8 hours weekly), achieving 99.9% pipeline reliability, and enabling elastic scaling for peak processing demands.
- Implemented comprehensive Azure Databricks data quality framework validating 200M+ daily records across claims, policy, and customer datasets, reducing data errors by 65% (from 15K to 5K monthly incidents), preventing \$2M+ in potential compliance issues, and improving downstream analytics accuracy from 82% to 96%.
- Orchestrated 100+ Azure Data Factory pipelines with Apache Airflow scheduling processing 3TB+ daily data across 20+ source systems, achieving 99.95% SLA compliance, automating incident recovery within 15 minutes, and reducing manual intervention by 70% through intelligent error handling and retry logic.
- Optimized Spark cluster performance through executor tuning, partition optimization, and caching strategies, reducing job execution time by 55% (from 4.5 hours to 2 hours), lowering compute costs by 42%, and improving resource utilization from 45% to 78% across 20-node Azure Databricks clusters.

Cummins

Pune, India

Data Modeler

01/2017 - 12/2018

- Designed 100+ normalized data models for manufacturing, supply chain, and inventory management systems using Erwin, reducing database redundancy by 40% (from 850GB to 510GB), improving query response times by 55% (from 45 seconds to 20 seconds), and supporting 20+ business intelligence applications serving 150+ stakeholders across global operations.
- Optimized Oracle database performance through strategic indexing, table partitioning, and query tuning, accelerating executive dashboard generation by 70% (from 30 minutes to 9 minutes), improving concurrent user capacity from 50 to 200 users, and reducing server resource consumption by 35%.
- Architected ETL workflows using IBM DataStage and Microsoft SSIS processing 2M+ daily records from 15+ heterogeneous source systems, improving data freshness from daily to hourly updates (95% latency reduction), enabling real-time inventory management, and supporting \$5B+ annual manufacturing operations.
- Implemented comprehensive metadata management framework documenting 500+ data elements with complete data lineage, business definitions, and transformation rules, reducing data discovery time by 60% (from 5 days to 2 days), accelerating impact analysis from 3 days to 4 hours, and improving cross-team data collaboration efficiency by 45%.
- Led 30+ data modeling review sessions with 12 business and technical stakeholders, establishing enterprise data standards and naming conventions, reducing modeling variations by 75%, accelerating new model deployment by 45% (from 4 weeks to 2.2 weeks), and achieving 98% stakeholder satisfaction scores.
- Developed 200+ Oracle database objects (tables, views, stored procedures, functions, triggers) following 3rd normal form principles, implementing robust referential integrity, optimizing join strategies, and creating comprehensive technical documentation supporting 30+ concurrent development projects.
- Engineered data lineage and impact analysis documentation covering 800+ data flows across supply chain, manufacturing, and financial systems, enabling proactive change management, reducing production incidents by 40%, and ensuring regulatory compliance for SOX audits with zero findings.

Professional Experience

Wide Business Solutions

Hyderabad, India

Project Engineer

03/2017 - 12/2017

- Deployed Python-based cloud infrastructure automation platform on Red Hat Linux enabling self-service resource provisioning, reducing manual deployment time by 80% (from 4 hours to 48 minutes), supporting 50+ development teams, and eliminating 90% of configuration errors through standardized templates.
- Engineered automated infrastructure testing workflows using Python CGI scripts and Selenium, reducing QA validation cycles by 60% (from 5 days to 2 days), accelerating release frequency from monthly to bi-weekly deployments, and improving code quality scores from 72% to 91%.
- Supported enterprise cloud migration project tracking 100+ deliverables using JIRA and MS Project, achieving 95% on-time delivery rate, maintaining zero budget overruns across \$2M project portfolio, and coordinating 8 cross-functional teams with 40+ members across development, QA, and operations.
- Collaborated with development, testing, and support teams ensuring seamless project transitions achieving 98% stakeholder satisfaction rating, resolving 200+ technical issues during deployment phases, and reducing production incident rate by 45% through comprehensive pre-deployment validation and automated smoke testing.

Technical Skills

Programming Languages: • Python • SQL • R • Java • Scala

Emerging Technologies: Applied AI • Machine Learning • LLM • NLP • Agentic AI • Generative AI • Cortex • Streamlit • Amazon Bedrock

Cloud Tools/ Pipeline Tools: • Microsoft Azure Synapse • Snowflake • DBT (Data Build Tool) • Databricks • AWS Services (Redshift • S3 • IAM • SQS • SNS • Lambda • EMR • Glue • Athena • AWS kinesis • Amazon EMR-Elastic MapReduce • CloudWatch)

Data Modeling Tools: • Erwin • Sparx EA • Star UML • Toad Data Modeler

Development Methodologies: • Agile • Scrum • Waterfall Model • UML • Design Patterns

IDEs Tools: • Jupyter Notebook • Anaconda • Hadoop • PyCharm • GitHub • Rally • JIRA

Databases : • MySQL • SQL Server • PostgreSQL • Oracle

NoSQL Databases: • MongoDB • Cassandra • HBase • Dynamo DB

ETL/Data-Pipeline Tools : • DataStage • Snap Logic • Spark • SSIS • Hive/Hadoop • Flink • PySpark • Apache Hadoop

Machine Learning : • TensorFlow • PyTorch • scikit-learn • Hugging Face • Regression • Classification • Dimensional Reduction • Ensemble Methods • Time Series Analysis

Statistics: • Probability Distributions • Descriptive Statistics • Statistical Learning • Inferential Statistics

Reporting Tools : • Tableau • Power BI • AWS Quick Sight

Education

William Jessup University

San Jose, United States

MS in Computer Science

KL University

Vijayawada, India

Bachelors in Mechanical Engineering